Komorebi Space

One realization, seen at four resolutions.

COHERENT VIEWS

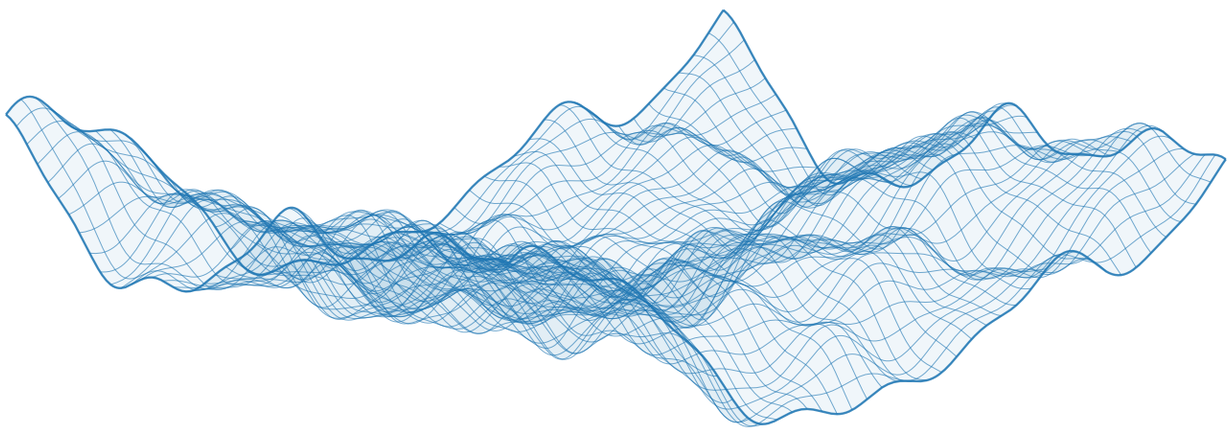
RETAINED DETAIL

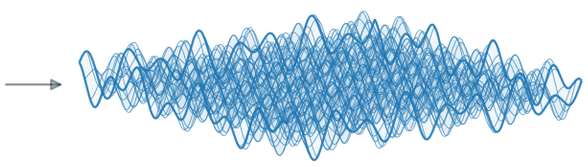
Detail height magnified 2×

x_3

$|k|_\infty \le 8$

$\dim X_3 = 289$





$d_2 = x_3 - J_2 x_2$

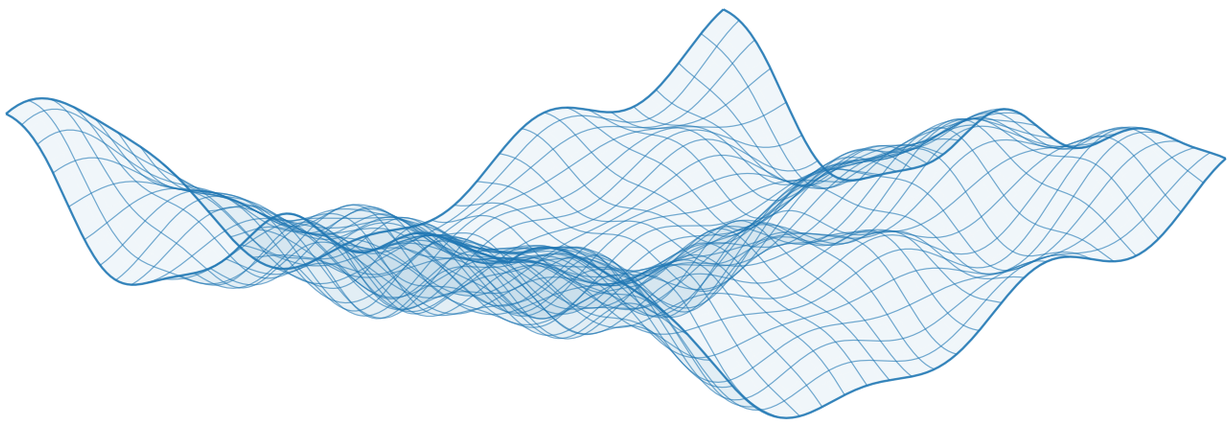
$P_2 d_2 = 0$

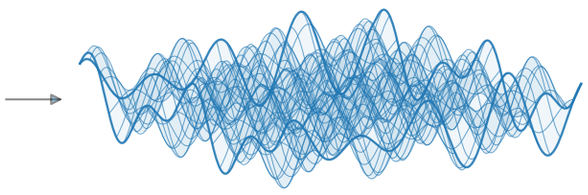
P_2

x_2

$|k|_\infty \le 4$

$\dim X_2 = 81$





$d_1 = x_2 - J_1 x_1$

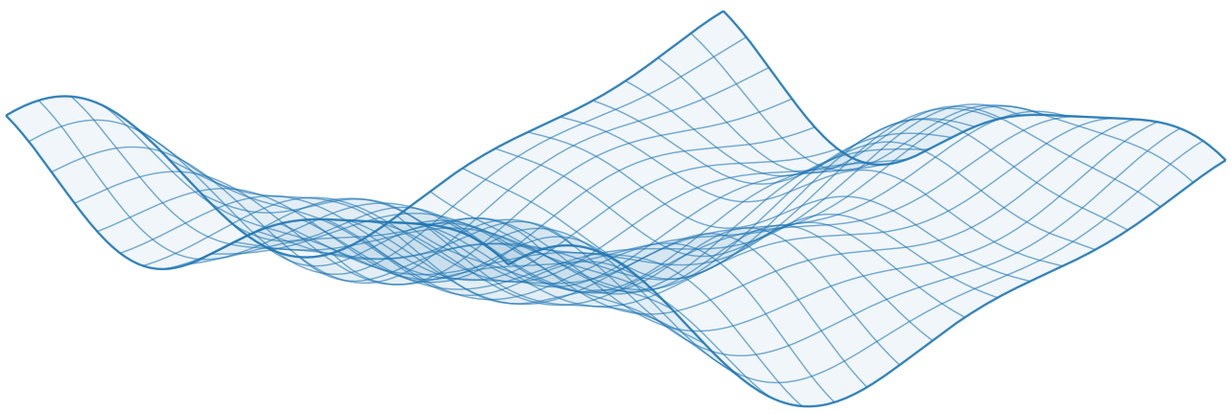
$P_1 d_1 = 0$

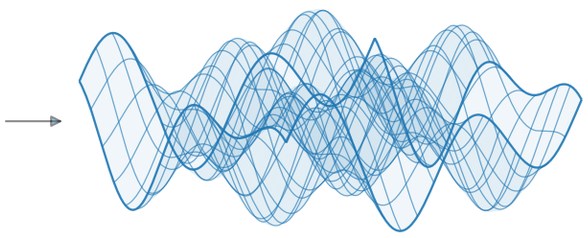
P_1

x_1

$|k|_\infty \le 2$

$\dim X_1 = 25$





$d_0 = x_1 - J_0 x_0$

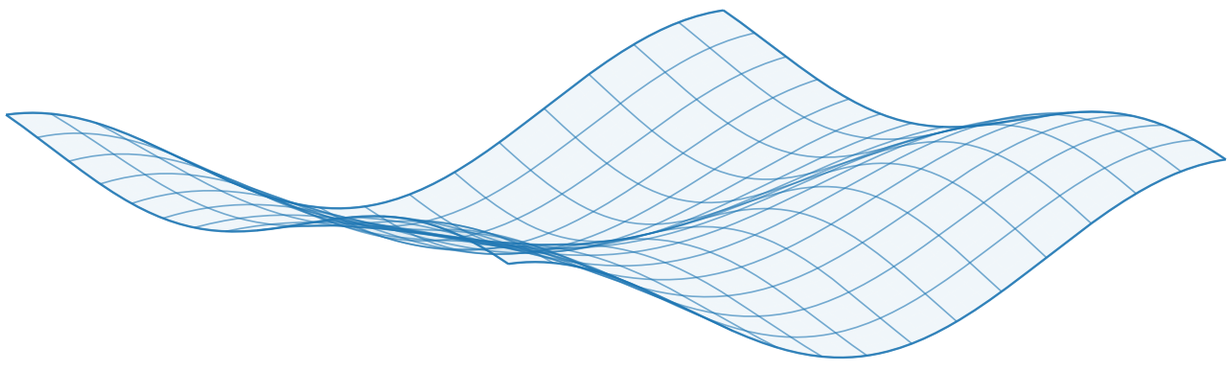
$P_0 d_0 = 0$

P_0

x_0

$|k|_\infty \le 1$

$\dim X_0 = 9$



Detail is information,

not inconsistency.

$P_n x_{n+1} = x_n$

Every restriction agrees exactly.

THE POINT IS THE WHOLE FAMILY

$$\mathcal{K}_3 = \{(x_0, x_1, x_2, x_3) \in \prod_{n=0}^{3} X_n \ : \ P_n x_{n+1} = x_n \quad (n = 0, 1, 2)\}$$

Each sheet is one Fourier function on the square torus, not the entire space X_n. Main sheets share one height scale.

Finite illustration of the realization construction. Geometry and identities are reproducible in Python.